



مدارس
للتنولوجيا
التطبيقية

IT Department
Web Programming

نماذج إختبارات مجمعة الصف الثالث شعبة البرمجة

Year's End

Units 17, 18, 19, 20, 21, 22, 23, 24

Exams

2025 - 2026

المسار الفني	مدارس تكنولوجيا التطبيقية
Program Name	فني تطوير مواقع وبرمجيات
المستهدف من التقييم	طلاب الصف الثالث
الوحدات التدريبية	Unit 17: React Unit 18: Node.JS Unit 19: PHP Unit 20: Advanced PHP Unit 21: Laravel Unit22 Advanced Python-2 Unit23 Design Patterns Unit24 Introduction to Software Testing and Capstone Project
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المعارف	Unit 17: React
	Writing asynchronous code using various techniques TPK20
	Basic routing and calling a controller method from a route TPK21
	Analyze and solve common web applications tasks by writing PHP programs TPK22
	Unit 18: Node.JS
	Database and data manipulation and Nodejs framework TPK19
	writing asynchronous code using various techniques TPK20
	Basic routing and calling a controller method from a route TPK21
	Analyse and solve common web applications tasks by writing PHP programs TPK22
	Unit 19: PHP
	Writing asynchronous code using various techniques TPK20
	Basic routing and calling a controller method from a route TPK21
	Analyze and solve common web applications tasks by writing PHP programs TPK22
	Unit 20: Advanced PHP
	Writing asynchronous code using various techniques TPK20
	Basic routing and calling a controller method from a route TPK21
	Analyze and solve common web applications tasks by writing PHP programs TPK22
	Unit 21: Laravel
	Analyze and solve common web applications tasks by writing PHP programs TPK22
	Unit22 Advanced Python-2
	Python and web GUI frameworks TPK17
	Data structures TPK23
	Algorithms TPK24
	Consol app TPK25
	Unit23 Design Patterns
	Digital tools options and uses TPK18
	Unit24 Introduction to Software Testing and Capstone Project
	software development TPK11
	software testing approaches (analytical, model based, methodical...etc) TPK12

The basics of agile testingAnd API testing				TPK13	
Unit 17	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK20	TPK21	TPK21	TPK21	TPK21
			TPK22	TPK22	TPK22
Unit 18	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK19	TPK19	TPK20	TPK20	TPK21
		TPK22	TPK19	TPK19	TPK22
Unit 19	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK20	TPK20	TPK21	TPK22	TPK22
Unit 20	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK20	TPK20	TPK21	TPK22	TPK22
Unit 21	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK22	TPK22	TPK22	TPK22	TPK22
Unit 22	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK17	TPK23	TPK23	TPK24	TPK25
Unit 23	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK18	TPK18	TPK18	TPK18	TPK18
Unit 24	Lesson one	Lesson two	Lesson three	Lesson four	Lesson Five
	TPK11	TPK12	TPK12	TPK13	TPK13

Exam 1

Group1: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
1	PHP is a client-side programming language.	False
2	PHP stands for "Hypertext Preprocessor."	True
3	PHP can be embedded inside HTML code.	True
4	Apache is a programming language used for building websites.	False
5	PHP scripts are executed on the server.	True
6	PHP variables must start with the \$ symbol.	True
7	Variable names in PHP are case-sensitive.	True
8	Multi-line comments in PHP use /* */.	True
9	A PHP function is a reusable block of code that performs a specific task.	True
10	The date() function can format and display the current date.	True

Group2: Dear Learner: Answer the following questions. (7 questions).

11	Question	What is the purpose of <code>php artisan make:model</code> ?
	Answer	Creates a new Eloquent model linked to a database table.
12	Question	What is Laravel Breeze?
	Answer	A lightweight authentication package using Tailwind CSS.
13	Question	What is the difference between PUT and PATCH?
	Answer	PUT replaces entire resource, PATCH updates partial data.
14	Question	How do you generate fake data in Laravel?
	Answer	Using model factories and seeders.
15	Question	What is the purpose of <code>@yield</code> in Blade?
	Answer	Defines a placeholder for content in a layout.
16	Question	What is the file extension for Laravel's Blade templates?
	Answer	.blade.php
17	Question	What are the four basic operations of CRUD?
	Answer	Create, Read, Update, Delete.

Group3: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
18	What is ReactJS primarily used for? a) Backend development b) Building user interfaces c) Database management d) Mobile app development	b
19	Which feature makes ReactJS efficient for UI updates? a) Virtual DOM b) Global CSS c) Server-side rendering d) Inline styles	a
20	What syntax extension does React use for templating? a) HTML b) CSS-in-JS c) JSX d) TypeScript	c
21	React components are: a) Non-reusable b) Reusable c) Only for class components d) Limited to 10 per app	b
22	To create a React app with Vite, run: a) npm create-react-app b) npm create vite@latest c) yarn add react d) npx install react	b
23	After creating a Vite project, you must run: a) npm run dev b) npm start c) npm build d) npm deploy	a
24	JSX stands for: a) JavaScript XML b) Java Syntax Extension	a

	c) JSON XML d) JS Extensions	
25	JSX allows embedding expressions with: a) Parentheses () b) Angle brackets <> c) Square brackets [] d) Curly braces {}	d
26	To render multiple JSX elements, wrap them in: a) <div> b) <Fragment> or <> c) d) <section>	b
27	React Router enables: a) Server-side routing b) State management c) Database queries d) Client-side navigation	d

Group4: Dear learner: Complete the question_(10 questions).

No	Question	Answer
28	React.js is primarily used for creating user interfaces for _____ applications.	Single-page (SPAs)
29	The main concept behind React's performance optimization is the _____ DOM.	Virtual
30	React components can be either _____ or class-based.	Functional
31	To manage state in functional components, you use the _____ Hook.	useState
32	The tool recommended for creating modern React apps is _____.	Vite
33	For complex state logic, the _____ hook is preferred over useState.	useReducer
34	Inline styles in React are written as _____ objects.	JavaScript
35	To install React Router, you use the package _____.	react-router-dom

36	A 404 page is implemented using the route path _____.	*
37	The _____ hook is used for side effects like data fetching.	useEffect

Group5: Match the correct answer from column A what suits column B (6 questions).

No	Column (A)	No.	Column (B)	Answer
38	GET	A	Create new data	38-B
39	POST	B	Retrieve data	39-A
40	PUT/PATCH	C	Delete data	40-D
41	DELETE	D	Update data	41-C
42	Laravel Breeze	E	Use Bootstrap 5	42-F
43	Laravel UI	F	Use Tailwind CSS	43-E

Group6: Dear Learner: Answer the following questions. (4 questions).

44	Question	What is the primary technical difference between TypeScript and JavaScript?
	Answer	TypeScript is statically typed, while JavaScript is dynamically typed.

45	Question	What is the purpose of the tsconfig.json file?
	Answer	To configure the behavior of the TypeScript compiler (defines compiler settings).

46	Question	What command do you use to compile a TypeScript file named app.ts?
	Answer	tsc app.ts

47	Question	What are the four main principles of Object-Oriented Programming supported by TypeScript?
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	Answer	Encapsulation, Inheritance, Polymorphism, Abstraction.
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Group7: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
48	The Agile model focuses on early and continuous delivery of software.	True
49	User Acceptance Testing (UAT) is performed by developers.	False
50	Unit testing checks how different modules work together.	False
51	System Integration Testing (SIT) ensures seamless communication between different systems.	True
52	Alpha testing is performed by real users in a production environment.	False
53	Black-box testing requires knowledge of the internal code structure.	False
54	Boundary Value Analysis (BVA) tests values at the edges of input ranges.	True
55	Decision table testing is used when there are multiple conditions and actions.	True
56	State transition testing is used when the system behaviour changes based on different states.	True
57	Checklist-based testing is best suited for regression testing.	True
58	Acceptance criteria define when a user story is considered complete.	True
59	Collaborative user story writing involves only developers, not testers.	False

Group8: Give the Scientific Name Questions: (7 questions).

No	Question	Answer
60	A type of operation where tasks can start and continue independently of each other is called:	Non-blocking
61	The tool that comes with Node.js for managing packages is the:	npm

62	A language that builds on JavaScript by adding static typing is called:	TypeScript
63	Functions that handle requests and responses between the client and server are known as:	Middleware
64	The acronym for an interface that connects different applications is:	API
65	The security mechanism that prevents web pages from making requests to a different domain is called:	CORS
66	The set of operations (Create, Read, Update, Delete) used to manage data is known as:	CRUD

Group9: Dear learner: Complete the question (6 questions).

No	Question	Answer
67	Testing software without predefined test cases is called _____ testing.	Exploratory
68	_____ testing checks individual code components in isolation.	Unit
69	_____ testing ensures different systems work together.	System Integration
70	_____ testing is done by real users before the final release.	Beta
71	A _____ describes a feature from a user's perspective.	User Story
72	_____ criteria define when a user story is complete.	Acceptance

Group10: Dear Learner: Answer the following questions (8 questions).

73	Question	What is the main purpose of arrow functions?
	Answer	To write short, simple one-line functions.
74	Question	What is the output of ceil(4.2)?
	Answer	5
75	Question	What is the main difference between GET and POST?
	Answer	GET sends data in the URL, POST sends data in the request body (hidden).
76	Question	Why should sensitive data not be sent using GET?
	Answer	Because it appears in the URL and can be stored in browser history or logs.

77	Question	What is the purpose of server-side validation?
	Answer	To ensure data accuracy and security even if client-side validation is bypassed.
78	Question	What does htmlspecialchars() prevent?
	Answer	Cross-Site Scripting (XSS) attacks.
79	Question	What does CRUD stand for?
	Answer	Create, Read, Update, Delete.
80	Question	What is a row in a database table?
	Answer	A single record (entry) representing one set of related data.

Group11: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
81	The command php artisan make:controller creates a new controller.	True
82	Middleware can restrict access to routes.	True
83	Blade files use the .blade.php extension.	True
84	Blade does not support loops and conditionals.	False
85	@extends is used to inherit a layout.	True
86	Migrations are used to modify database structure.	True
87	CSRF protection is enabled by default in Laravel forms.	True
88	API resources format JSON responses.	True
89	Form validation in Laravel only checks for empty fields.	False
90	The php artisan route:list command displays all routes.	True

Exam 2

Group1: Dear Learner: Answer the following questions. (10 questions):

1	Question	What is software testing?
	Answer	Ensuring a product works as required and is free of bugs.
2	Question	Why is testing necessary?
	Answer	To find bugs early, improve quality, reduce costs, and meet user expectations.
3	Question	What is the difference between verification and validation?
	Answer	Verification checks if the software meets requirements; validation checks if it meets user needs.
4	Question	What is the difference between testing and debugging?
	Answer	Testing finds bugs; debugging fixes them.
5	Question	What is defect clustering?
	Answer	Most bugs are found in a few specific areas of the software.
6	Question	What is the pesticide paradox in testing?
	Answer	Repeating the same tests stops finding new bugs; fresh tests are needed.
7	Question	Name two test activities in the STLC.
	Answer	Test planning and test execution.
8	Question	What is testware?
	Answer	All artifacts (documents, scripts, tools) produced during testing.
9	Question	What is the role of a test analyst?
	Answer	Designs test cases based on requirements.
10	Question	What is shift-left testing?
	Answer	Performing testing earlier in the development lifecycle.

Group2: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement. (10 questions):

No	Question	Answer
11	The \$_FILES superglobal array is used to access information about an uploaded file.	True
12	The move_uploaded_file() function is used to delete a file from the server.	False
13	PDO stands for PHP Data Orientation.	False
14	PDO can only be used to connect to MySQL databases.	False
15	Prepared statements in PDO help prevent SQL injection attacks.	True
16	Cookies are stored on the server.	False
17	To delete a cookie, you set its expiration time to a past date.	True
18	Sessions are stored on the client's browser.	False
19	The session_start() function is used to begin a new session or resume an existing one.	True
20	The session_unset() function removes all session variables.	True

Group3: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
21	What does PHP stand for? a) Personal Hosting Program b) Hypertext Preprocessor c) Programming Hypertext Pages d) Page Hosting Processor	b
22	PHP code is written inside: a) <php> tags b) <script> tags c) <?php ?> tags d) {} braces	c
23	Which symbol is used to declare variables in PHP? a) @ b) \$	b

	c) % d) &	
24	Which statement is faster for output? a) echo b) print c) console.log d) display	a
25	Which operator is used for string concatenation in PHP? a) + b) & c) . d) ,	c
26	Single-line comments in PHP use: a) <!-- --> b) // c) \$ d) /* */	b
27	Which operator assigns values to variables? a) Arithmetic b) Assignment c) Comparison d) Logical	b
28	The break keyword in PHP: a) Skips current iteration b) Stops the loop c) Repeats the loop d) Ends PHP script	b
29	The continue keyword in PHP: a) Ends the loop b) Skips current iteration c) Exits PHP code d) Repeats the loop	b
30	Which operator increments a variable by 1? a) ++ b) --	a

c) +=	
d) =+	

Group4: Dear learner: Give signific term for question (8 questions).

No	Question	Answer
31	The virtual representation of the actual DOM that React uses for efficient updates.	Virtual DOM
32	The syntax extension for JavaScript that allows writing HTML-like code in React.	JSX
33	The modern build tool that's faster than Create React App for setting up React projects.	Vite
34	The self-contained, reusable building blocks of a React application.	Components
35	The type of component defined as a JavaScript function.	Functional component
36	The type of component that uses ES6 classes and extends React.Component.	Class component
37	The React object that holds mutable data specific to a component.	state
38	The immutable data passed from parent to child components.	props

Group5: Dear learner: Choose the correct answer (7 questions).

No	Question	Answer
39	What is the extension for Blade files? a) .php b) .blade c) .blade.php d) .html.php	c
40	How do you escape output in Blade? a) {!! \$var !!} b) {{ \$var }}	b

	c) @escape(\$var) d) @{{{ \$var }}}	
41	Which directive extends a layout in Blade? a) @include b) @extends c) @yield d) @section	b
42	What are migrations used for? a) Storing session data b) Modifying database structure c) Rendering views d) Handling HTTP requests	b
43	Which command creates a migration? a) php artisan make:migration b) php artisan create:migration c) php artisan new:migration d) php migrate:make	a
44	What is the purpose of seeders? a) Generate fake data b) Define routes c) Create controllers d) Compile assets	a
45	How is CSRF protection added to forms in Laravel? a) @protect b) @secure c) @form_token d) @csrf	d

Group6: Match each term with its correct definition (6 questions).

No	Term	No	Definition	Answer
46	Testing	A	Fixing identified defects in the code.	46-B
47	Debugging	B	Detecting defects in the software.	47-A
48	Verification	C	Ensuring the software meets user needs.	48-D
49	Validation	D	Checking if the software meets specified requirements.	49-C
50	Error	E	A mistake made by a developer.	50-E
51	Defect	F	A flaw in the software causing incorrect behavior.	51-F

Group7: Dear Learner: Answer the following questions (10 questions).

52	Question	What is the main goal of Agile methodology?
	Answer	Deliver value quickly through iterative development and customer collaboration.
53	Question	Name two Agile frameworks.
	Answer	Scrum and Kanban.
54	Question	What is a Sprint in Agile?
	Answer	A short development cycle (1-4 weeks) where a set of tasks is completed.
55	Question	What is the purpose of a Daily Scrum?
	Answer	To provide quick updates on progress and blockers.
56	Question	What does a Product Owner do in Scrum?
	Answer	Prioritizes tasks and manages the product backlog.
57	Question	Why is API testing important?
	Answer	It ensures APIs function correctly, securely, and efficiently.
58	Question	Name two common data formats used in API testing.
	Answer	JSON and XML.

59	Question	What is the difference between unit and integration testing in APIs?
	Answer	Unit tests check individual components, while integration tests check API interactions.
60	Question	What tool is used for testing REST APIs?
	Answer	Postman or SoapUI.
61	Question	What does security testing in APIs involve?
	Answer	Identifying vulnerabilities like SQL injection or unauthorized access.

Group8: Give the Scientific Name Questions: (7 questions).

No	Question	Answer
62	The part of a website that represents the browser on the computer is called the:	client
63	The server or place that contains the backend code is called the:	Server
64	The open-source, cross-platform runtime environment for executing JavaScript code is:	Node.js
65	The open-source JavaScript and Web Assembly engine developed by Google is called:	V8
66	A Node.js framework known for its simple design that lets developers create customized applications is:	Express.js
67	A library that facilitates real-time communication by allowing clients and servers to exchange messages instantly is:	Socket.io
68	A type of operation where a task must finish before the next one can start is called:	Blocking

Group9: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
69	Laravel follows the MVC architecture.	True

70	MVC stands for Model, View, Controller.	True
71	Laravel does not enforce a separation between business logic and presentation.	False
72	Models in Laravel handle database interactions.	True
73	Views in Laravel are responsible for business logic.	False
74	Controllers act as an intermediary between Models and Views.	True
75	The default Laravel welcome page runs on http://localhost:8000.	True
76	Blade templates are stored in resources/views/.	True
77	The routes/ folder defines API and web routes.	True
78	Route parameters are defined using {} in Laravel.	True

Group10: Dear learner: Complete the question (6 questions).

No	Question	Answer
79	The process of ensuring software is free of bugs and works as expected is called _____.	Software Testing
80	Checking if the software meets specified requirements is known as _____.	Verification
81	A set of steps to verify a feature's functionality is called a _____.	Test Case
82	Testing individual units of code is known as _____.	Unit Testing
83	The model that delivers software in small, usable increments is the _____ model.	Incremental
84	The tool used to track and manage bugs is called a _____.	Bug Tracking Tool

Group11: Match the correct answer from column A what suits column B (6 questions).

No	Column (A)	No	Column (B)	Answer
85	<BrowserRouter>	A	Hook to extract dynamic URL parameters	85 → C
86	<Routes>	B	Client-side navigation component	86 → E
87	<Route>	C	Top-level component for client-side routing	87 → F
88	<Link>	D	Placeholder for nested route content	88 → B
89	useParams()	E	Container for <Route> definitions	89 → A
90	<Outlet />	F	Maps a URL path to a component	90 → D

Exam 3

Group1: Dear Learner: Answer the following questions. (10 questions).

1	Question	What is ReactJS?
	Answer	A JavaScript library for building user interfaces using reusable components.
2	Question	What is JSX?
	Answer	A syntax extension for JavaScript that allows HTML-like code in JS files.
3	Question	What is the Virtual DOM?
	Answer	A lightweight copy of the real DOM for efficient updates.
4	Question	How does React handle rendering?
	Answer	It updates only the parts of the DOM that change (via Virtual DOM diffing).
5	Question	What are React components?
	Answer	Reusable UI building blocks (like custom HTML elements).
6	Question	What are props?
	Answer	Immutable data passed from parent to child components.
7	Question	How do you pass props to a component?
	Answer	<code><ChildComponent name="Yosra" age={30} /></code> .
8	Question	What is Vite?
	Answer	A fast build tool for modern React apps.
9	Question	How do you create a React app with Vite?
	Answer	Run <code>npm create vite@latest</code> and select React.
10	Question	How do you style React components?
	Answer	Inline styles, CSS modules, or libraries like Tailwind.

Group2: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (7 questions).

No	Question	Answer
11	The count() function returns the number of elements in an array.	True
12	include() stops the execution of the script if the file is missing.	False
13	require() throws a fatal error if the file is missing.	True
14	include_once() ensures that a file is included only once in the script.	True
15	The \$_POST method sends data through the URL.	False
16	GET requests can be bookmarked because the data is visible in the URL.	True
17	The isset() function can be used to check if a cookie has been set.	True

Group3: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
18	XAMPP includes: a) PHP, Apache, MySQL b) HTML, CSS, JS c) Python, Django, SQLite d) Windows Server	a
19	Which keyword is used to define a user function in PHP? a) method b) function c) define d) var	b
20	Which array function adds an element at the beginning? a) array_push() b) array_pop() c) array_shift() d) array_unshift()	d

21	In associative arrays, values are stored with: a) Keys b) Indexes only c) Strings only d) None	a
22	Which is NOT a built-in numeric function in PHP? a) pow() b) sqrt() c) strlen() d) abs()	c
23	Which type of array can contain other arrays? a) Indexed array b) Associative array c) Multidimensional array d) String array	c
24	Which function throws a fatal error if the file is missing? a) require() b) include() c) echo() d) isset()	a
25	To prevent duplicate file inclusion, you use: a) repeat() b) require() c) include_once() d) addfile()	c
26	Which CRUD operation retrieves data? a) INSERT b) UPDATE c) SELECT d) DELETE	c
27	Which CRUD operation removes records? a) DELETE b) DROP c) REMOVE d) TRUNCATE	a

Group4: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
28	Laravel follows the MVC architecture.	True
29	MVC stands for Model, View, Controller.	True
30	Laravel does not enforce a separation between business logic and presentation.	False
31	Models in Laravel handle database interactions.	True
32	Views in Laravel are responsible for business logic.	False
33	Controllers act as an intermediary between Models and Views.	True
34	The default Laravel welcome page runs on http://localhost:8000.	True
35	Blade templates are stored in resources/views/.	True
36	The routes/ folder defines API and web routes.	True
37	Route parameters are defined using {} in Laravel.	True

Group5: Dear Learner: Answer the following questions (3 questions).

38	Question	What is the main difference between GET and POST?
	Answer	GET sends data in the URL, POST sends data in the request body (hidden).

39	Question	Why should sensitive data not be sent using GET?
	Answer	Because it appears in the URL and can be stored in browser history or logs.

40	Question	What does CRUD stand for?
	Answer	Create, Read, Update, Delete.

Group6: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
41	What is the main goal of software testing? a) To write code b) To ensure software works as expected and is bug-free	b

	c) To design user interfaces d) To manage project budgets	
42	Which of the following is NOT a benefit of software testing? a) Early bug detection b) Increased development costs c) Improved software quality d) Meeting user expectations	b
43	Verification in software testing checks: a) If the software meets user needs b) Only the GUI design c) If the software meets specified requirements d) Only performance issues	c
44	What is the difference between testing and debugging? a) Testing fixes bugs; debugging finds them b) Testing finds bugs; debugging fixes them c) Both are the same d) Testing is done after debugging	b
45	Which principle states that most bugs are found in a few specific modules? a) Defect clustering b) Early testing c) Pesticide paradox d) Exhaustive testing	a
46	Which testing type checks individual units or components of code? a) Integration testing b) System testing c) Unit testing d) Acceptance testing	c
47	What does the "pesticide paradox" imply in testing? a) Repeating the same tests finds new bugs b) New tests are needed to find new bugs c) Bugs are like pests that never go away d) Automated tests are better than manual tests	b

48	Which phase of STLC involves creating test cases? a) Requirement Analysis b) Test Planning c) Test Case Design d) Test Execution	c
49	Which of the following is a non-functional test? a) Unit testing b) Performance testing c) Integration testing d) Regression testing	b
50	What is the purpose of regression testing? a) To test new features only b) To replace unit testing c) To test only the user interface d) To ensure new changes don't break existing functionality	d

Group7: Match the function with its action (6 questions):

No	Function	No	action	Answer
51	setcookie()	A	Starts a new or resumes an existing session.	51→C
52	session_start()	B	Moves an uploaded file to a new location.	52→A
53	move_uploaded_file()	C	Creates or modifies a cookie.	53→B
54	session_destroy()	D	Removes all session variables.	54→E
55	session_unset()	E	Completely destroys all data registered to a session.	55→D
56	isset()	F	Checks if a variable is declared and is not null.	56→F

Group8: Test Code Debugging Challenge (7 questions):

57	Question	What's wrong with this migration? Schema::create('products', function (Blueprint \$table) { \$table->id(); \$table->string('name'); \$table->timestamps(); \$table->decimal('price',8,2); });
	Answer	Fields should be added before timestamps(): Schema::create('products', function (Blueprint \$table) { \$table->id(); \$table->string('name'); \$table->decimal('price',8,2); // Before timestamps \$table->timestamps(); });
58	Question	Fix this route? Route::get('/user', 'UserController@show');
	Answer	Route::get('/user', [UserController::class, 'show']);
59	Question	Why won't this update work? <form action="/users/{ {{ \$user->id }} " method="POST"> @csrf <!-- form fields --> </form>
	Answer	Missing @method('PUT'): <form action="/users/{ {{ \$user->id }} " method="POST"> @csrf @method('PUT') <!-- Add this --> <!-- form fields --> </form>
60	Question	Fix this validation: \$request->validate(['email' => 'required email']);
	Answer	Add unique rule for new records: \$request->validate(['email' => 'required email unique:users,email]);

61	Question	Find and fix the error. @foreach(\$users as \$user) {{ \$user.name }} @endforeach
	Answer	Wrong property access syntax. Fixed: @foreach(\$users as \$user) {{ \$user->name }} // Use -> not . @endforeach

62	Question	Why is this form vulnerable? <form method="POST" action="/login"> <!-- form fields --> </form>
	Answer	Missing @csrf: <form method="POST" action="/login"> @csrf <!-- Add this --> <!-- form fields --> </form>

63	Question	Fix this redirect to send flash message: return redirect('dashboard');
	Answer	Add flash message: return redirect('dashboard')->with('success', 'User created!');

Group9: Match each SDLC model with its description (6 questions).

No	Model	No	Description	Answer
64	Waterfall	A	Delivers parts of the software quickly for early user feedback.	64-C
65	V-Shaped	B	Combines iterative and incremental approaches, focusing on adaptability.	65-E
66	Iterative	C	Follows strict sequential phases with heavy documentation.	66-D
67	Incremental	D	Repeats cycles of development and testing to refine the product.	67-A
68	Agile	E	Emphasizes testing at every development phase.	68-B
69	Spiral	F	Focuses on risk analysis and iterative refinement (not mentioned but plausible for advanced testing).	69-F

Group10: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
70	Agile testing starts only after development is complete.	False
71	Daily Scrums are optional in Agile teams.	False
72	Kanban uses fixed sprints like Scrum.	False
73	The Product Owner in Scrum sets task priorities.	True
74	API testing helps detect bugs early in development.	True
75	Security testing is not part of API testing.	False
76	Load testing simulates a single user accessing an API.	False
77	Unit testing checks how different APIs work together.	False
78	JSON and XML are common data formats in API testing.	True
79	API testing reduces costs by catching errors early.	True

Group11: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
80	What is the main goal of software testing? a) To write code b) To ensure software works as expected and is bug-free c) To design user interfaces d) To manage project budgets	b
81	Which of the following is NOT a benefit of software testing? a) Early bug detection b) Increased development costs c) Improved software quality d) Meeting user expectations	b
82	Verification in software testing checks: a) If the software meets user needs b) Only the GUI design c) If the software meets specified requirements d) Only performance issues	c
83	What is the difference between testing and debugging? a) Testing fixes bugs; debugging finds them b) Testing finds bugs; debugging fixes them	b

	c) Both are the same d) Testing is done after debugging	
84	Which principle states that most bugs are found in a few specific modules? a) Defect clustering b) Early testing c) Pesticide paradox d) Exhaustive testing	a
85	Which testing type checks individual units or components of code? a) Integration testing b) System testing c) Unit testing d) Acceptance testing	c
86	What does the "pesticide paradox" imply in testing? a) Repeating the same tests finds new bugs b) New tests are needed to find new bugs c) Bugs are like pests that never go away d) Automated tests are better than manual tests	b
87	Which phase of STLC involves creating test cases? a) Requirement Analysis b) Test Planning c) Test Case Design d) Test Execution	c
88	Which of the following is a non-functional test? a) Unit testing b) Performance testing c) Integration testing d) Regression testing	b
89	What is the purpose of regression testing? a) To test new features only b) To replace unit testing c) To test only the user interface d) To ensure new changes don't break existing functionality	d

Exam 4

Group1: Give the Scientific Name Questions (8 questions):

No	Question	Answer
1	What design pattern does Laravel use to separate logic, UI, and processes?	MVC
2	Which Laravel tool helps manage packages and dependencies?	Composer
3	What is the name of Laravel's command-line tool for generating code and managing tasks?	Artisan
4	What is the syntax used in Blade to output a variable safely?	{{ }}
5	What is the term for a gatekeeper that checks or modifies requests before they reach a route?	Middleware
6	What is the term for version-controlled files that modify database tables?	Migrations
7	What is Laravel's templating engine called?	Blade
8	What is the process of defining user permissions called?	Authorization

Group2: Dear learner: Choose the correct answer (12 questions).

No	Question	Answer
9	What is Node.js? a) A JavaScript framework b) A new programming language c) A runtime environment for executing JavaScript code d) A web browser	c

10	How would you describe a "Non-Blocking" operation? a) Tasks must complete in order, one after the other. b) Tasks can start and run independently without waiting. c) It prevents other operations from starting. d) It is another term for a synchronous operation.	b
11	How do you verify the version of Node.js installed on your system? a) node -v b) npm -v c) install node d) run node	a
12	The fs (file system) module in Node.js is an example of what? a) A Third-Party Module b) A Local Module c) A Core Module d) A Framework	c
13	Which command is used to install TypeScript globally using npm? A) npm install typescript B) npm add -g typescript C) node install typescript D) npm install -g typescript	d
14	What is the correct file extension for a TypeScript source file? A) .ts B) .txt C) .js D) .typescript	a
15	What is the primary purpose of Express.js? a) To style web pages b) To build server-side web applications and APIs c) To manage client-side state d) To create database schemas	b
16	Which command is used to initialize a new Node.js project and create a package.json file?	d

	a) npm start b) node init c) npm install express d) npm init	
17	Which of the following is NOT a feature of Express.js mentioned in the text? a) Middleware b) Template Engines c) Virtual DOM d) Error Handling	c
18	How is a specific employee's ID typically accessed from the URL in an Express route like /employees/:id? a) req.query.id b) req.body.id c) req.params.id d) req.url.id	c
19	What does CRUD stand for? a) Create, Read, Update, Delete b) Call, Render, Update, Destroy c) Create, Render, Use, Deploy d) Calculate, Return, Use, Define	a
20	The main purpose of CORS is to: A) Prevent all access to a web server. B) Securely allow web pages to make requests to a different domain. C) Encrypt data stored in a database. D) Improve the performance of template rendering.	b

Group3: Dear Learner: Answer the following questions. (10 questions):

21	Question	What is the main goal of a retrospective in Agile?
	Answer	To reflect on what went well and identify improvements.
22	Question	What is a test case?
	Answer	A set of steps, conditions, and expected results to verify a feature.

23	Question	What is the difference between a test case and a test scenario?
	Answer	A test case is detailed; a test scenario is broader (e.g., "Check login" vs. "Enter username & password").
24	Question	What is a defect in software testing?
	Answer	An error causing the software to behave incorrectly.
25	Question	Name two common types of bugs.
	Answer	Syntax errors and logic errors.
26	Question	What is a bug tracking tool?
	Answer	Software used to record, track, and manage defects (e.g., Jira).
27	Question	What is the Waterfall model?
	Answer	A sequential SDLC model where each phase must be completed before moving to the next.
28	Question	What is Agile testing?
	Answer	Testing in short iterations with continuous feedback and adaptation.
29	Question	What is regression testing?
	Answer	Re-running old tests to ensure new changes don't break existing features.
30	Question	What is the purpose of UAT (User Acceptance Testing)?
	Answer	To confirm the software meets business/user needs before release.

Group4: Match the HTTP method to the CRUD operation it typically performs (6 questions):

No	HTTP Method	No	CRUD Operation	Answer
31	GET	A	Delete	31-B
32	POST	B	Read	32-C
33	PUT	C	Create	33-E
34	PATCH	D	Update (Partial)	34-D
35	DELETE	E	Update (Full)	35-A
36	All of them	F	CRUD	36-F

Group5: Dear learner: Give signific term for question (7 questions).

No	Question	Answer
37	The React Hook that manages state in functional components.	useState
38	The utility-first CSS framework that can be configured with Vite.	Tailwind CSS
39	The library that enables client-side navigation in React applications.	React Router
40	The top-level component that syncs your app with the browser's URL.	BrowserRouter
41	The component that defines the mapping between URLs and components.	Route
42	The container component that holds multiple Route definitions.	Routes
43	The component used for navigation that prevents full page reloads.	Link

Group6: Dear learner: Choose the correct answer (8 questions).

No	Question	Answer
44	What is Laravel? a) A JavaScript framework b) A PHP framework c) A CSS library d) A database system	b
45	Which architecture does Laravel follow? a) MVP b) MVC c) MVVM d) Flux	b
46	What does Laravel's Eloquent ORM do? a) Handles HTTP requests b) Compiles CSS	c

	c) Manages database interactions d) Renders Blade templates	
47	Which tool is used to manage Laravel dependencies? a) npm b) Composer c) Yarn d) Pip	b
48	What command starts the Laravel development server? a) php artisan start b) php artisan serve c) npm run dev d) composer serve	b
49	What is the purpose of middleware? a) Style web pages b) Filter HTTP requests c) Seed databases d) Generate fake data	b
50	Which method returns a view in Laravel? a) return view() b) return html() c) return blade() d) return render()	a
51	What does Route::resource() generate? a) A single route b) CRUD routes c) API endpoints d) Authentication routes	b

Group7: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
52	Software testing ensures that a product is free of bugs and performs as required.	False
53	Early testing helps reduce costs by catching bugs before they escalate.	True
54	The pesticide paradox suggests that repeating the same tests always finds new bugs.	False
55	Testing is the same for all types of software (e.g., games vs. banking apps).	False
56	Regression testing ensures that new changes don't break existing functionality.	True
57	Unit testing checks how different software components work together.	False
58	Performance testing checks how fast and stable software is under different workloads.	True
59	Shift-Left Testing means delaying testing until the end of development.	False
60	A retrospective meeting focuses only on what went wrong in a project.	False
61	A test case includes steps, expected results, and actual results.	True

Group8: Dear Learner: Answer the following questions : (6 questions).

62	Question	What are the three main parts of a website's architecture?
	Answer	The client, the server, and the database.

63	Question	Define: Node.js?
	Answer	Node.js is an open-source, cross-platform runtime environment for executing JavaScript code outside a web browser.

64	Question	What is the key difference between blocking and non-blocking operations?
	Answer	Answer: Blocking operations must be completed before the next task can start, while non-blocking operations can run independently without waiting.

65	Question	What command do you use to check if Node.js is installed correctly on your system?
	Answer	node -v

66	Question	What is the primary purpose of Express.js?
	Answer	To simplify the process of building web applications and APIs in Node.js.

67	Question	What is the role of middleware?
	Answer	To handle and process requests and responses between the client and the server.

Group9: Dear learner: Complete the question. (7 questions):

No	Question	Answer
68	PHP variables start with the symbol _____.	\$
69	PHP comments that span multiple lines use _____.	/* */
70	PHP constants are defined using the function _____.	define()
71	The default extension for PHP files is _____.	.php
72	To add two variables in PHP, we use the _____ operator.	+
73	The output function in PHP faster than print is _____.	Echo
74	PHP scripts run on the _____ side.	Server

Group10: Match PHP Functions with Behavior (6 questions):

No	Functions	No	Behavior	Answer
75	include()	A	Converts special characters to HTML entities	75 → c
76	require()	B	Includes file, fatal error if missing	76 → b
77	include_once()	C	Includes file, warning if missing	77 → d
78	require_once()	D	Includes file only once, warning if missing	78 → f
79	htmlspecialchars()	E	Connects PHP to MySQL	79 → a
80	mysqli_connect()	F	Includes file only once, fatal error if missing	80 → e

Group11: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
81	ReactJS is a back-end JavaScript framework.	False
82	React was developed by Facebook.	True
83	React is primarily used for building single-page applications.	True
84	React uses a virtual DOM to improve performance.	True
85	React compares the virtual DOM with the real DOM to determine changes.	True
86	The virtual DOM updates the entire real DOM on every state change.	False
87	JSX is executed directly by browsers.	False
88	The useReducer hook takes a reducer function and an initial state value as arguments.	True
89	When using useReducer, you call a dispatch function with an action object to update the state.	True
90	The useContext hook is primarily used for managing state in deeply nested components without prop drilling.	True

Exam 5

Group1: Dear Learner: Answer the following questions : (4 questions).

1	Question	What is the key structural difference between SQL and NoSQL databases?
	Answer	SQL uses tables with fixed rows and columns, while NoSQL uses flexible documents or other models.

2	Question	What is the main benefit of using a template engine like EJS?
	Answer	It separates the presentation layer (HTML) from the application logic (JavaScript).

3	Question	What command is used to start an Express server with automatic restarts on file changes?
	Answer	nodemon server.js

4	Question	What is Crud?
	Answer	The set of operations (Create, Read, Update, Delete) used to manage data

Group2: Give the Scientific Name Questions (7 questions):

No	Question	Answer
5	A variable type that holds sequences of characters.	String
6	A PHP statement that outputs content and is faster than print.	echo
7	A PHP data type that represents true or false.	Boolean
8	A PHP statement that stops a loop.	break
9	A PHP statement that skips the current iteration.	continue
10	A block of code that can be reused multiple times in PHP is called a.	Function
11	The PHP function that shows the current date is.	date()

Group3: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
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12	useEffect is a React Hook that lets you perform side effects in class components.	False
13	Fetching data from an API is an example of a side effect that useEffect can handle.	True
14	Link components prevent full page reloads.	True
15	useParams can access route parameters.	True
16	Nested routes require the Outlet component.	True
17	You can embed JavaScript expressions in JSX using curly braces {}.	True
18	JSX fragments are written as <>...</>.	True
19	React components must always start with a capital letter.	True
20	Props are immutable within child components.	True
21	Hooks can only be used in functional components.	True

Group4: Give the Scientific Name Questions (8 questions):.

No	Question	Answer
22	What design pattern does Laravel use to separate logic, UI, and processes?	MVC
23	Which Laravel tool helps manage packages and dependencies?	Composer
24	What is the name of Laravel's command-line tool for generating code and managing tasks?	Artisan
25	What is the syntax used in Blade to output a variable safely?	{{ }}
26	What is the term for a gatekeeper that checks or modifies requests before they reach a route?	Middleware
27	What is the term for version-controlled files that modify database tables?	Migrations
28	What is Laravel's templating engine called?	Blade

29	What is the process of defining user permissions called?	Authorization
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Group5: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
30	What does Shift-Left Testing emphasize? a) Delaying testing until deployment b) Testing early in the development cycle c) Only manual testing d) Skipping unit tests	b
31	Which SDLC model delivers software in increments? a) Waterfall b) V-Model c) Incremental d) Spiral	c
32	What is the key focus of Agile methodology? a) Rigid planning b) Early and continuous delivery c) No customer involvement d) Avoiding testing	b
33	Which testing is done without predefined test cases? a) Regression testing b) Exploratory testing c) Unit testing d) System testing	b
34	What is the main purpose of a retrospective meeting? a) To assign blame for bugs b) To plan the next project c) To reflect on improvements d) To skip testing	c
35	Which component is NOT part of a test case? a) Test Case ID b) Expected Result c) Developer's salary d) Test Steps	c

36	A bug that causes incorrect results without crashing is called: a) Syntax error b) Runtime error c) Logic error d) Security bug	c
37	What does CI/CD stand for in DevOps? a) Continuous Inspection/Continuous Deployment b) Continuous Integration/Continuous Delivery c) Code Inspection/Code Deployment d) Customer Input/Customer Delivery	b
38	Which SDLC model is sequential and non-iterative? a) Agile b) Waterfall c) Spiral d) Iterative	b
39	Which testing checks software compatibility across devices? a) Performance testing b) Compatibility testing c) Regression testing d) Unit testing	b

Group6: Match the API Testing Type to its Purpose (6 questions).

No	Testing Type	No	Purpose	Answer
40	Unit Testing	A	Simulates multiple users accessing the API simultaneously.	40 → B
41	Integration Testing	B	Tests individual API components in isolation.	41 → C
42	Performance Testing	C	Checks how different APIs interact with each other.	42 → D
43	Load Testing	D	Evaluates speed, efficiency, and scalability.	43 → A

44	Security Testing	E	Protects against external threats and data breaches.	44 → E
45	Validation Testing	F	Verifies if the API meets its intended specifications.	45 → F

Group7: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
46	Node.js itself is a programming language.	False
47	The V8 engine was developed by Google.	True
48	Node.js is considered a framework like Express.js.	False
49	When you install Node.js, NPM is installed automatically.	True
50	The http module is an example of a third-party module.	False
51	TypeScript code is executed directly by a web browser's JavaScript engine.	False
52	The private access modifier allows a class property to be accessed from anywhere in your code.	False
53	Inheritance allows a subclass to inherit properties and methods from a superclass.	True
54	Middleware functions handle requests and responses between the client and server.	True
55	A PUT request is used specifically for deleting records in a REST API.	False
56	The app.use(express.json()) middleware is necessary to parse JSON data from the request body.	True
57	The req.body object is used to access data sent in the request's JSON body.	True

Group8: Give the Scientific Name Questions (6 questions).

No	Question	Answer
58	A set of five design principles that make object-oriented software more maintainable and extensible.	SOLID
59	A pattern that defines how objects communicate and distribute responsibilities.	Behavioral Design Pattern

60	A design pattern where the algorithm structure is defined in a base class with customizable steps.	Template Method Pattern
61	Ability to use existing code structure across different subclasses.	Reusability
62	Modifying specific steps while keeping the general algorithm intact.	Customization
63	Ability of code to adapt to changes in behavior.	Flexibility

Group9: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
64	In OOP, an object is a template for a class.	False
65	The \$this keyword is available outside of a class's methods.	False
66	A public property can be accessed from anywhere.	True
67	A private property can be accessed by a class derived from the parent class.	False
68	Encapsulation is the OOP principle for restricting access to internal details via access modifiers	True
69	Inheritance allows a class to acquire properties and methods from another class.	True
70	The json_encode() function converts a JSON string into a PHP variable.	False
71	A Try-Catch block is used for handling errors gracefully.	True
72	In the example, divide(10, 0) will execute without causing an error.	False
73	The enctype attribute for a file upload form must be set to multipart/form-data.	True

Group10: Give the Scientific Name Questions (10 questions).

No	Question	Answer
74	A function that is automatically called when you create an object from a class.	constructor
75	The keyword used to refer to the current object inside a class method.	\$this

76	The access modifier that allows a property to be accessed from anywhere in the code.	public
77	The programming paradigm that is about creating objects containing both data and functions.	OOP
78	The principle of restricting direct access to some of an object's components.	encapsulation
79	An instance of a class.	object
80	A template for objects, defining properties and methods.	class
81	The PHP superglobal array that contains information about files uploaded via an HTTP POST method.	\$_FILES
82	A lightweight format for storing and exchanging data.	JSON
83	A code block that acts as a safety net for handling runtime errors.	Try-Catch

Group11: Give the Scientific Name Questions: (7 questions).

No	Question	Answer
84	A type of operation where tasks can start and continue independently of each other is called:	Non-blocking
85	The tool that comes with Node.js for managing packages is the:	npm
86	A language that builds on JavaScript by adding static typing is called:	TypeScript
87	Functions that handle requests and responses between the client and server are known as:	Middleware
88	The acronym for an interface that connects different applications is:	API
89	The security mechanism that prevents web pages from making requests to a different domain is called:	CORS
90	The set of operations (Create, Read, Update, Delete) used to manage data is known as:	CRUD

Exam 6

Group1: Give the Scientific Name Questions (8 questions).

No	Question	Answer
1	An array that stores values using numeric indexes is a.	Indexed Array
2	An array that uses named keys to store values is.	Associative Array
3	The function that counts array elements is.	count()
4	A PHP function that includes a file but continues execution even if missing.	include()
5	A PHP function that includes a file but throws a fatal error if missing.	require()
6	A PHP superglobal that collects form data sent via URL parameters.	\$_GET
7	A PHP superglobal that collects form data sent via HTTP POST.	\$_POST

Group2: Complete the Sentence Questions (8 questions):

No	Question	Answer
8	The date() function is used to format a local _____ and time.	date
9	The _____ method of a PDO object is used to create a prepared statement.	prepare
10	You should use the _____ function to check if a cookie is set before trying to use its value.	isset
11	Session variables are stored on the _____, while cookies are stored on the _____.	server, client
12	The server identifies a user's session on subsequent requests using a unique _____ that is typically stored in a cookie.	session ID
13	In the PDO DSN for MySQL, you specify the database driver using _____:.	mysql

14	The <code>json_encode()</code> function converts a PHP <u> </u> or array into a JSON string.	object
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Group3: Match the PHP Superglobal with its purpose (6 questions):

No	Superglobal	No	purpose	Answer
15	<code>\$_FILES</code>	A	Stores variables passed to the script via HTTP POST.	15→B
16	<code>\$_SESSION</code>	B	Stores information about uploaded files.	16→D
17	<code>\$_COOKIE</code>	C	Stores variables passed to the script via HTTP GET.	17→E
18	<code>\$_POST</code>	D	Stores session variables for the current user.	18→A
19	<code>\$_GET</code>	E	Stores variables passed to the script via HTTP cookies.	19→C
20	<code>\$_SERVER</code>	F	Stores information about headers, paths, and script locations.	20→F

Group4: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
21	To define routes, use: a) <code><Routes></code> and <code><Route></code> b) <code><Link></code> and <code><NavLink></code> c) <code><div></code> and <code></code> d) <code><BrowserRouter></code> only	a
22	A 404 page uses the route path: a) <code>"/404"</code> b) <code>"*"</code> c) <code>"/error"</code> d) <code>"not-found"</code>	b
23	Inline styles in JSX use: a) CSS strings b) HTML classes c) JSON files d) JavaScript objects	d

24	To install Tailwind, run: a) npm install tailwindcss b) npm install react-tailwind c) yarn add bootstrap d) npx create-tailwind	a
25	To share state between components: a) Use global variables b) Copy-paste code c) Lift state up d) Avoid props	c
26	Props are used to: a) Modify parent state b) Pass data to children c) Replace Redux d) Style components	b
27	Prop drilling refers to: a) Passing props through multiple components b) Using useEffect c) Styling components d) Server-side rendering	a
28	Rules of Hooks include: a) Call Hooks in loops b) Call Hooks at the top level c) Use only in class components d) Avoid useState	b
29	useReducer is used for: a) Simple state b) Styling c) Complex state logic d) Prop drilling	c
30	useContext helps with: a) Global state management b) Local state only c) API calls d) Inline CSS	a

Group5: Match each test case element with its description (6 questions).

No	Component	No	Description	Answer
31	Test Case ID	A	Detailed steps to execute the test.	31-C
32	Description	B	What should happen if the feature works correctly.	32-F
33	Preconditions	C	Unique identifier for the test case.	33-E
34	Test Steps	D	Input values used in the test.	34-A
35	Expected Result	E	Requirements before the test can run.	35-B
36	Test Data	F	What feature or function is being tested?	36-D

Group6: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
37	What is the primary focus of Agile methodology? a) Strict documentation b) Long development cycles c) Avoiding all changes d) Customer satisfaction and flexibility	d
38	Which Agile framework uses fixed-length sprints? a) Scrum b) Kanban c) Extreme Programming (XP) d) Waterfall	a
39	What is the main purpose of a Sprint Retrospective? a) To demo new features b) To improve team processes c) To assign new tasks d) To write documentation	b
40	Which role removes obstacles in Scrum? a) Product Owner b) Scrum Master c) Developer d) Stakeholder	b

41	Kanban uses: a) A visual board (To Do, In Progress, Done) b) Fixed sprints c) Only written reports d) No task tracking	a
42	What does load testing simulate? a) A single API request b) Only manual testing c) Multiple users accessing an API d) Database queries	c
43	Which data formats are used in API testing? a) HTML and CSS b) XML and JSON c) PDF and DOCX d) MP3 and MP4	b
44	What is the main benefit of early bug detection in API testing? a) Saves time and money b) Increases costs c) Delays releases d) Requires more documentation	a
45	What does integration testing check? a) Single API components b) How different APIs work together c) Only security flaws d) UI responsiveness	b
46	Which testing ensures an API meets its intended purpose? a) Performance testing b) Load testing c) Validation testing d) Security testing	c

Group7: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
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47	SOLID principles are used in many design patterns.	True
48	The Open-Closed Principle states that a class should be open for extension and closed for modification.	True
49	Behavioral design patterns focus on object creation.	False
50	Structural design patterns are about organizing classes and objects into larger structures.	True
51	A single responsibility principle suggests that a class should have only one reason to change.	True
52	Creational patterns focus on class instantiation and object creation.	True
53	Design patterns speed up development by providing proven Solutions.	True
54	Design patterns are compared to Lego bricks because both help in building complex structures from smaller reusable units.	True
55	Creational design patterns are mainly concerned with how objects interact with each other.	False
56	Factory Pattern is used to ensure only one instance of a class exists.	False
57	Some design patterns can be combined to solve more complex problems.	True
58	The Singleton pattern ensures multiple instances of a class can exist at the same time	False

Group8: Dear learner: Choose the correct answer (7 questions).

No.	Question	Answer
59	Which of the following is NOT one of the SOLID principles? a) Single Responsibility Principle b) Open-Closed Principle c) Dependency Creation Principle d) Liskov Substitution Principle	C
60	Which design pattern category deals with communication between objects? a) Structural	B

	b) Behavioral c) Creational d) Functional	
61	The main goal of the Interface Segregation Principle is: a) To increase inheritance b) To split large interfaces into smaller ones c) To combine interfaces d) To remove interfaces completely	B
62	Which of the following is the main task of design patterns? a) Provide high-level guidelines b) Provide reusable solutions to common design problems. c) Focus only on database logic d) Eliminate OOP concepts	B
63	Which of the following is a Creational Design Pattern? a) Adapter b) Singleton c) Composite d) Proxy	B
64	The main goal of Creational design patterns is: a) Code execution speed b) Object creation flexibility c) Database optimization d) User interface design	B
65	Structural patterns are mainly concerned with: a) Data storage b) Object creation c) Organizing relationships between objects d) Error handling	C

Group9: Dear Learner: Answer the following questions (7 questions).

66	Question	What is PHP mainly used for?
	Answer	Server-side web development.
67	Question	What is the purpose of Apache in PHP setup?
	Answer	It serves web pages to users.
68	Question	Write an example of a PHP variable.
	Answer	\$name = "Ali";
69	Question	How do you define a constant in PHP?
	Answer	define("PI", 3.14);
70	Question	What does the continue statement do in a loop?
	Answer	Skips current iteration.
71	Question	What is the difference between built-in and user-defined functions?
	Answer	Built-in functions are provided by PHP; user-defined functions are created by developers.
72	Question	What does the pow(2, 3) function return?
	Answer	8

Group10: Dear learner: Give signific term question (8 questions).

NO	Question	Answer
73	A utility-first CSS framework for rapid UI development.	Tailwind CSS
74	The library that enables client-side navigation in React applications.	React Router
75	The top-level component that syncs your app with the browser's URL.	BrowserRouter
76	The component that defines the mapping between URLs and components.	Route
77	The container component that holds multiple Route definitions.	Routes
78	The component used for navigation that prevents full page reloads.	Link
79	The special Link component that applies styling when active.	NavLink
80	The hook that extracts dynamic parameters from the URL.	useParams

Group11: Match the PDO-related term with its description (6 questions):

No	term	No	description	Answer
81	DSN	A	The method used to send parameters to a prepared statement.	81→C
82	Prepared Statement	B	A value passed to setAttribute() to control how errors are handled.	82→D
83	execute()	C	A connection string that defines the database driver and details.	83→A
84	PDO::ATTR_ERRMODE	D	A pre-compiled SQL query that separates logic from data.	84→B
85	fetch()	E	The method used to retrieve a result row from a query.	85→E
86	PDOException	F	The type of exception thrown by PDO on errors.	86→F

Exam 7

Group1 Dear learner: Choose the correct answer (8 questions):

No	Question	Answer
1	Which function is crucial for moving an uploaded file to a permanent location? a) copy_file() b) move_uploaded_file() c) save_upload() d) store_file()	b
2	Which HTML form attribute is essential for enabling file uploads? a) method="GET" b) action="upload.php" c) enctype="multipart/form-data" d) type="file"	c
3	What is the primary advantage of using PDO over the older mysql extension? a) It's faster b) It provides a consistent interface for multiple databases c) It doesn't require a password d) It only works with MySQL	b
4	Which method is used to execute a prepared statement in PDO? a) run() b) execute() c) query() d) fetch()	b
5	What is the main security benefit of using prepared statements? a) They make queries run faster b) They prevent SQL injection attacks c) They compress data d) They encrypt database connections	b

6	Which function is used to create a cookie in PHP? a) create_cookie() b) make_cookie() c) setcookie() d) set_cookie()	c
7	Where is session data stored by default? a) On the client's browser b) In a database c) In a cookie d) On the server	d
8	Which function must always be called before using \$_SESSION variables? a) session_begin() b) start_session() c) session_start() d) session_init()	c

Group2: Dear learner: Find the error in the code (6 questions).

No.	Question	Answer
9	<?php \$name = "Ali" echo \$name; ?>	Error: Missing semicolon after ` \$name = "Ali" <b>Fix:</b> ` \$name = "Ali";`
10	<?php \$1name = "Ali"; echo \$1name; ?>	Error: Variable name cannot start with a number Fix: \$name1 = "Ali";
11	function greet(\$name) { echo "Hello " \$name; }	Error: Missing dot (.) for concatenation. Fix: echo "Hello " . \$name;
12	\$nums = array("1","2","3"); echo array_pop(\$nums, "4");	Error: array_pop() takes only one parameter . Fix: echo array_pop(\$nums);
13	echo strlen[“Hello”];	Error: Wrong brackets . Fix: echo strlen("Hello");

14	<code>\$arr = array("name"=>"Ali", "age"=>20); echo \$arr[0];</code>	Error: Associative arrays require keys , not indexes. Fix: <code>echo \$arr["name"];</code>
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Group3: Dear Learner: Answer the following questions (7 questions).

15	Question	What is state in React?
	Answer	Data that changes over time and affects rendering.
16	Question	How do you declare state in functional components?
	Answer	<code>const [count, setCount] = useState(0);</code> .
17	Question	What is useReducer used for?
	Answer	Managing complex state logic with actions/reducers.
18	Question	What is the purpose of useEffect?
	Answer	To perform side effects (e.g., API calls) in components.
19	Question	What is useContext?
	Answer	A hook for accessing context values without prop drilling.
20	Question	What is prop drilling?
	Answer	Passing props through multiple layers of components.
21	Question	What is Tailwind CSS?
	Answer	A utility-first CSS framework for rapid UI development.

Group4: Match the correct answer from column A what suits column B (6 questions).

No	Column (A)	No	Column (B)	Answer
21	Virtual DOM	A	Hook for complex state logic with reducers	21 → C
22	JSX	B	Hook for side effects (e.g., data fetching)	22 → E
23	useState	C	A. Lightweight in-memory representation of the actual DOM	23 → F

24	useEffect	D	Hook for accessing React context globally	24 → B
25	useReducer	E	HTML-like syntax extension for JavaScript	25 → A
26	useContext	F	Hook for managing state in functional components	26 → D

Group5: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
27	Structural design patterns deal with how objects are created	False
28	The Adapter pattern helps make two incompatible interfaces work together.	True
29	The Composite pattern is mainly used to represent a tree structure.	True
30	Creational design patterns cannot be reused in different projects.	False
31	Singleton provides centralized control for shared resources	True
32	Creational design patterns focus on how objects are created rather than how they work.	True
33	Structural patterns increase flexibility and adaptability in applications.	True
34	Behavioral design patterns are concerned with how objects interact with each other	True
35	Behavioral design patterns do not improve testability of code.	False
36	Template Method reduces code duplication by reusing common logic	True
37	Template Method pattern helps in creating reusable and customizable algorithms.	True
38	Behavioral patterns increase flexibility of applications.	True

Group6: Match the Node.js component with its description (6 questions).

No	Component	No	Description	Answer
39	Client	A	The place that contains the backend code.	39-C
40	Server	B	Contains all the data that must be saved.	40-A
41	Database	C	Represents the browser on the computer.	41-B
42	V8 Engine	D	A core component that compiles JavaScript code.	42-D
43	Express.js	E	A library for real-time client-server communication.	43-F
44	Socket.io	F	A lightweight framework for building web apps.	44-E

Group7: Dear learner: Put "True" in front of the incorrect statement (7 questions).

No	Question	Answer
45	A Graphical User Interface (GUI) requires users to have deep programming knowledge to interact with electronic devices.	False
46	Buttons are clickable elements that trigger actions within a GUI.	True
47	PyQt is considered the easiest Python GUI library for beginners because it comes installed with Python.	False
48	The geometry() method in Tkinter sets the window's starting size and position.	True
49	The pack() method is a geometry manager used to position widgets in a window.	True
50	Tkinter's default look and feel is considered modern and sleek.	False
51	Tkinter.ttk is a modern set of widgets that automatically matches the style of the user's operating system.	True

Group8: Dear Learner: Answer the following questions (4 questions).

52	Question	What are Creational Patterns?
	Answer	Creational patterns focus on flexible and controlled object creation.

53	Question	What are Structural Patterns?
	Answer	Structural patterns focus on organizing classes and objects into larger, maintainable structures.

54	Question	Explain the main purpose of Behavioral Design Patterns.
	Answer	Behavioral design patterns define how objects communicate, distribute responsibilities, and collaborate, improving flexibility and scalability of applications.

55	Question	Discuss the benefits of using the Template Method pattern in large projects.
	Answer	It reduces code duplication, promotes reusability, and enforces a consistent algorithm structure. It allows flexibility through subclass customization while keeping the base algorithm stable.

Group9: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
56	Which of the following is NOT a common GUI element mentioned in the text? a. Text boxes b. Windows c. Command lines d. Scroll bars	c
57	According to the text, which Python GUI library is best for creating simple, basic programs and is easy to start with? a. Kivy	d

	b. PyQt c. PyGTK d. Tkinter	
58	What is the main advantage of using Tkinter.ttk over standard Tkinter? a. It is better for mobile apps. b. It provides more advanced widgets. c. It automatically matches the look of the user's operating system. d. It is easier to customize colors.	c
59	What is the purpose of the title() method in a Tkinter window? a. To set the size of the window. b. To change the text at the top of the window. c. To set the background color. d. To set the minimum size of the window.	b
60	What is the root widget in a Tkinter application? a. A type of button. b. The main window or parent window. c. A field for text input. d. An image file.	b
61	Which Python GUI library is described as "The Mobile One" and is best for apps with touchscreens? a. Tkinter b. PyQt c. Kivy d. wxPython	c
62	The geometry() method string format widthxheight+X+Y defines what for a window? a. Width and height only. b. Width, height, and title. c. Width, height, and starting position from the top-left corner. d. Width, height, and minimum size.	c

63	What is the purpose of the pack() method when used on a widget in Tkinter? a. To set the widget's color. b. To change the widget's font. c. To run the app's main event loop. d. To place the widget on the current window.	d
64	What is the disadvantage of using Tkinter mentioned in the text? a. It does not work on macOS. b. It requires an external library to be installed. c. Its default look and feel are dated. d. It is not compatible with other Python tools.	c
65	Which Python GUI library is recommended for making complex, professional applications that need to look native on different operating systems? a. Tkinter b. Kivy c. PyQt d. Pygame	c

Group10: Match the GUI element to its description (6 questions).

No	GUI Element	No	Description	Answer
66	Icons	A	Clickable elements that trigger actions.	66-D
67	Buttons	B	Fields for inputting text.	67-A
68	Text boxes	C	Rectangular areas on the screen that display information.	68-B
69	Scroll bars	D	Small pictures that represent actions or objects.	69-E
70	Windows	E	Used to navigate through large amounts of content.	70-C
71	Menu	F	Lists of options that can be selected.	71-F

Group11: Complete the Sentence Questions (10 questions).

No	Question	Answer
72	In design patterns classification, the two criteria are _____ and _____.	Purpose, Scope
73	The _____ Pattern allows you to define a family of algorithms and switch between them at runtime.	Strategy
74	A charger plug converter is an example of the _____ Pattern.	Adapter
75	The three main types of design patterns are _____, _____, and _____.	Creational, Structural, Behavioral
76	The Factory Pattern provides a common _____ for creating objects of different subclasses.	Interface
77	----- patterns focus on how objects are created .	Creational design
78	The ----- pattern helps to make incompatible interfaces compatible.	Adapter
79	----- pattern can improve performance by saving memory.	Singleton
80	----- patterns are concerned with relationships between objects.	Structural
81	The----- Method pattern provides a uniform interface for creating objects.	Factory

Group12: Match the correct answer from column A what suits column B (8 questions).

No	Column (A)	No.	Column (B)	Answer
82	Singleton	A	Converts one interface to another	82→F
83	Factory Method	B	Example of Factory Method Usage	83→D
84	Adapter	C	Example of Adapter Usage	84→A
85	Composite	D	Creates objects without exposing creation logic	85→H
86	Database Manager	E	Example of Singleton Usage	86→E
87	Car Factory	F	Ensures one instance only	87→B
88	HDMI to AV Converter	G	Example of Composite Usage	88→C
89	File System	H	Represents tree-like structures	89→G

Exam 8

Group1: Dear learner: Choose the correct answer (8 questions):

NO	Question	Answer
1	What is stored on the client's browser to identify a user's session? a) All their session variables b) A session ID c) Their username d) A unique token	b
2	Which of these is a key difference between sessions and cookies? a) Sessions are stored on the client, cookies on the server. b) Cookies are stored on the client, sessions on the server. c) Sessions are always persistent. d) Cookies cannot be deleted.	b
3	Which function checks if a variable is set and is not NULL? a) is_null() b) check() c) isset() d) empty()	c
4	Which PDO method is best for executing a SELECT statement with variables? a) exec() b) query() c) prepare() and execute() d) run()	c
5	Which PHP function converts a PHP array or object into a JSON string? a) json_convert() b) json_string() c) json_decode() d) json_encode()	d

6	Which filter would you use to check if a value is a whole number? a) FILTER_SANITIZE_NUMBER_INT b) FILTER_VALIDATE_INT c) FILTER_VALIDATE_FLOAT d) FILTER_SANITIZE_SPECIAL_CHARS	b
7	To validate multiple input variables at once, which function should you use? a) filter_var() b) filter_input() c) filter_var_array() d) array_filter()	c
8	Which PDOException is caught in the database connection example? a) A general Exception b) A file reading error c) A database connection error d) A division by zero error	c

Group2: Dear Learner: Answer the following questions (5 questions).

9	Question	What is React Router?
	Answer	A library for client-side navigation in SPAs.
10	Question	How do you define a route in React Router?
	Answer	<Route path="/about" element={<About />} />.
11	Question	What is <Link> used for?
	Answer	Client-side navigation without page reloads.
12	Question	How do you access URL parameters?
	Answer	const { id } = useParams();.
13	Question	How do you install Bootstrap in React?
	Answer	npm install bootstrap and import its CSS.

Group3: Match the correct answer from column A what suits column B (6 questions).

No	Column (A)	No.	Column (B)	Answer
14	@yield	A	Extends a layout	14-C
15	@section	B	Includes another view	15-E
16	@extends	C	Defines a placeholder in a layout	16-A
17	@include	D	Outputs escaped data	17-B
18	{{ }}	E	Defines a content block	18-D
19	@if	F	Conditional rendering	19-F

Group4: Dear learner: Complete the question. (7 questions):

No	Question	Answer
20	PHP variables start with the symbol _____.	\$
21	PHP comments that span multiple lines use _____.	/* */
22	PHP constants are defined using the function _____.	define()
23	The default extension for PHP files is _____.	.php
24	To add two variables in PHP, we use the _____ operator.	+
25	The output function in PHP faster than print is _____.	echo
26	PHP scripts run on the _____ side.	Server

Group5: Dear learner: Choose the correct answer (7 questions).

No.	Question	Answer
27	Which of these is a Structural Design Pattern? a) Singleton b) Factory Method c) Adapter d) Prototype	C

28	What does the Adapter pattern achieve? a) Creates new objects b) Makes incompatible interfaces work together c) Provides only one instance d) Hides internal code	B
29	Using many design patterns in one project can lead to: a) Cleaner code b) Complexity c) Better performance d) Faster development	B
30	Which of the following provides centralized control? a) Adapter b) Singleton c) Composite d) Factory	B
31	Which pattern is best for creating multiple product types in an e-commerce system? a) Factory Method b) Singleton c) Adapter d) Composite	A
32	Which type of design pattern focuses on object interaction and responsibility distribution? a) Creational b) Structural c) Behavioral d) Architectural	C
33	Separation of Concerns means: a) Using fewer classes b) Dividing code into distinct responsibilities c) Writing longer methods d) Increasing memory	B

Group6: Match the term with its correct type (6 questions).

No	Term	No	Type	Answer
34	Node.js	A	Framework	34-C
35	JavaScript	B	Engine	35-D
36	Express.js	C	Runtime Environment	36-A
37	V8	D	Programming Language	37-B
38	http	E	Third Party Module	38-F
39	mysql	F	Core Module	39-E

Group7: Dear learner: Give The Correct Scientific Term (7 questions).

No	Question	Answer
40	This is a Tkinter geometry manager used to pack or place a widget on the current window.	pack()
41	This is the main window of a Tkinter application, which must be created before any other windows.	Root widget
42	This is the method used to set the starting size and position of a Tkinter window.	geometry()
43	This is the standard Python library that provides a simple and intuitive way to create graphical user interfaces.	Tkinter
44	This is a Tkinter widget that can be used to display either text or images.	Label
44	This specific method is used to change settings of a Tkinter window, such as its background color.	configure()
45	This is the part of a Tkinter application that handles mouse and keyboard inputs and communicates with the operating system.	Main event loop

Group8: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (9 questions):

No	Question	Answer
46	A Checkbutton's state is stored in an external control variable.	True
47	The Frame widget can be used to organize GUI elements.	True
48	The Spinbox widget only supports floating-point values.	False
49	The Scale widget is often used for volume controls in multimedia apps.	True
50	A Combobox is a widget that presents its options as a scrollable list that is open by default.	False
51	The Listbox widget supports the selection of multiple items at once.	True
52	The pack geometry manager is the only way to arrange a Frame widget in a Tkinter window.	False
53	The <<ComboboxSelected>> event is triggered when the selected item in a Combobox changes.	True
54	The insert() function is used to add items to a Listbox.	True

Group9: Dear learner: Choose the correct answer (9 questions).

No	Question	Answer
55	Which widget is the simplest in Tkinter and is used for displaying a single line of text? a. Button b. Entry c. Label d. Messagebox	c
56	Which of the following is not a valid horizontal text alignment option for the anchor parameter? a. tk.W b. tk.N c. tk.E d. "center"	b
57	What is the purpose of the command keyword argument in a Button widget? a. To define the button's text.	c

	b. To change the button's background color. c. To specify the function to be called when the button is clicked. d. To set the button's height and width.	
58	The tkinter.messagebox module is primarily used for: a. Creating buttons with images. b. Displaying pop-up windows for information, warnings, or confirmation. c. Arranging widgets within a window. d. Changing the font of a Label.	b
59	How do you change the background color of a Button when the mouse cursor is hovering over it? a. bg parameter b. bd parameter c. activebackground parameter d. foreground parameter	c
60	What must you do to use functions from the tkinter.messagebox module? a. Nothing, it's included with tkinter. b. You must import it separately with from tkinter import messagebox. c. You must import it with import tkinter.messagebox. d. Both b and c are standard import practices.	d
61	What is geometry manager in Tkinter that is used to organize widgets in blocks before placing them in the parent widget? a. config() b. pack() c. anchor d. command	b
62	Which parameter is used to display an image on a Button instead of text? a. bg b. image c. photo	b

	d. display	
63	The askyesno function from tkinter.messagebox is a type of message box that: a. Provides an informational message. b. Displays an error. c. Asks the user for a yes/no confirmation. d. Displays a warning.	c

Group10: Match Array Functions with their Purpose (6 questions):

No	Functions	No.	Purpose	Answer
64	count()	A	Removes element from start	64→ f
65	array_push()	B	Reverses array order	65→ c
66	array_pop()	C	Adds element at end	66→ e
67	array_shift()	D	Adds element at start	67→ a
68	array_unshift()	E	Removes element from end	68→ d
69	array_reverse()	F	Number of elements in array	69→ b

Group11: Dear Learner: Answer the following questions. (10 questions).

70	Question	What is Laravel?
	Answer	A free, open-source PHP framework for building web applications.
71	Question	What does MVC stand for in Laravel?
	Answer	Model, View, Controller.
72	Question	What command starts a Laravel development server?
	Answer	<code>php artisan serve.</code>
73	Question	Where are Laravel routes defined?
	Answer	In <code>routes/web.php</code> (for web) and <code>routes/api.php</code> (for APIs).
74	Question	What is Blade in Laravel?
	Answer	Laravel's templating engine for dynamic HTML rendering.
75	Question	What is Eloquent ORM?

	Answer	Laravel's database interaction system using PHP models instead of SQL.
76	Question	What is a migration in Laravel?
	Answer	A file that defines database table changes (create, modify, delete).
77	Question	How do you create a new controller?
	Answer	<code>php artisan make:controller ControllerName.</code>
78	Question	What is middleware in Laravel?
	Answer	A filter that checks/modifies HTTP requests before they reach routes/controllers.
79	Question	What is CSRF protection in Laravel?
	Answer	Security feature preventing cross-site request forgery attacks.

Group12: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (6 questions):

No	Question	Answer
80	Debugging is the process of finding bugs, while testing is the process of fixing them.	False
81	Quality Assurance (QA) focuses on preventing bugs before they occur.	True
82	DevOps combines development and operations to deliver software faster.	True
83	An array can only store one type of value in PHP.	False
84	Associative arrays use numeric indexes only.	False
85	A multidimensional array contains one or more arrays inside it.	True

Group13: Dear learner: Complete the question (6 questions).

No	Question	Answer
86	The process of ensuring software is free of bugs and works as expected is called _____.	Software Testing

87	Checking if the software meets specified requirements is known as _____.	Verification
88	A set of steps to verify a feature's functionality is called a _____.	Test Case
89	Testing individual units of code is known as _____.	Unit Testing
90	The model that delivers software in small, usable increments is the _____ model.	Incremental
91	The tool used to track and manage bugs is called a _____.	Bug Tracking Tool

Exam 9

Group1 Dear Learner: Answer the following questions. (7 questions).

1	Question	What is a GUI and what does it replace?
	Answer	A GUI (Graphical User Interface) allows users to interact with electronic devices using visual elements like icons and buttons, replacing the need for complex command-line interfaces.
2	Question	What are three examples of common GUI elements?
	Answer	Common GUI elements include buttons, menus, and text boxes.
3	Question	Why is Tkinter considered a good choice for beginners in Python GUI development?
	Answer	Tkinter is simple to learn and comes pre-installed with Python, making it easy to start with GUI development.
4	Question	In Tkinter, what is the purpose of the root widget?
	Answer	The root widget is the main or parent window of a Tkinter application, and all other widgets must be created within it.
5	Question	What are two ways to load images into a Tkinter label?
	Answer	You can use the PhotoImage class to load GIF, PGM, PPM, and PNG images, or use the Pillow library for other image formats.
6	Question	Briefly describe the purpose of the <u>command</u> parameter in a Tkinter Button widget.
	Answer	The command parameter is used to specify the name of the Python function or method that should be called whenever the button is clicked.
7	Question	What is the primary function of the <code>tkinter.messagebox</code> module?

	Answer	Its primary function is to display ready-made pop-up message boxes to the user to provide information, warnings, errors, or ask for confirmation.
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Group2: Put "True" in front of the correct statement and "False" in front of the incorrect statement (7 questions).

No	Question	Answer
8	A Graphical User Interface (GUI) requires users to have deep programming knowledge to interact with electronic devices.	False
9	Buttons are clickable elements that trigger actions within a GUI.	True
10	PyQt is considered the easiest Python GUI library for beginners because it comes installed with Python.	False
11	The geometry() method in Tkinter sets the window's starting size and position.	True
12	The pack() method is a geometry manager used to position widgets in a window.	True
13	Tkinter's default look and feel is considered modern and sleek.	False
14	Tkinter.ttk is a modern set of widgets that automatically matches the style of the user's operating system.	True

Group3: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
15	Which of the following is NOT a common GUI element mentioned in the text? a. Text boxes b. Windows c. Command lines d. Scroll bars	c
16	According to the text, which Python GUI library is best for creating simple, basic programs and is easy to start with? a. Kivy	d

	b. PyQt c. PyGTK d. Tkinter	
17	What is the main advantage of using Tkinter.ttk over standard Tkinter? a. It is better for mobile apps. b. It provides more advanced widgets. c. It automatically matches the look of the user's operating system. d. It is easier to customize colors.	c
18	What is the purpose of the title() method in a Tkinter window? a. To set the size of the window. b. To change the text at the top of the window. c. To set the background color. d. To set the minimum size of the window.	b
19	What is the root widget in a Tkinter application? a. A type of button. b. The main window or parent window. c. A field for text input. d. An image file.	b
20	Which Python GUI library is described as "The Mobile One" and is best for apps with touchscreens? a. Tkinter b. PyQt c. Kivy d. wxPython	c
21	The geometry() method string format widthxheight+X+Y defines what for a window? a. Width and height only. b. Width, height, and title. c. Width, height, and starting position from the top-left corner. d. Width, height, and minimum size.	c
22	What is the purpose of the pack() method when used on a widget in Tkinter?	d

	a. To set the widget's color. b. To change the widget's font. c. To run the app's main event loop. d. To place the widget on the current window.	
23	What is the disadvantage of using Tkinter mentioned in the text? a. It does not work on macOS. b. It requires an external library to be installed. c. Its default look and feel are dated. d. It is not compatible with other Python tools.	c
24	Which Python GUI library is recommended for making complex, professional applications that need to look native on different operating systems? a. Tkinter b. Kivy c. PyQt d. Pygame	c

Group4: Match the Agile Testing Life Cycle Phase to its Description (6 questions).

No	Phase	No	Description	Answer
25	Impact Assessment	A	Daily meetings to update testing status and set priorities.	25 → C
26	Test Planning	B	Final checks to decide if features are ready for release.	26 → D
27	Daily Scrums	C	Gathering stakeholder input for future implementation.	27 → A
28	Test Agility Review	D	Preparing a testing schedule and expected outcomes.	28 → E
29	Release Readiness	E	Evaluating progress to ensure alignment with the plan.	29 → B
30	Security Testing	F	Conducting security audits before deployment.	30 → F

Group5: Dear learner: Give The Correct Scientific Term (7 questions).

No	Question	Answer
31	This is a Tkinter geometry manager used to pack or place a widget on the current window.	pack()
32	This is the main window of a Tkinter application, which must be created before any other windows.	Root widget
33	This is the method used to set the starting size and position of a Tkinter window.	geometry()
34	This is the standard Python library that provides a simple and intuitive way to create graphical user interfaces.	Tkinter
35	This is a Tkinter widget that can be used to display either text or images.	Label
36	This specific method is used to change settings of a Tkinter window, such as its background color.	configure()
37	This is the part of a Tkinter application that handles mouse and keyboard inputs and communicates with the operating system.	Main event loop

Group6: Match the testing level to its description (6 questions).

No	Testing Level	No	Description	Answer
38	Unit Testing	A	Tests interactions between systems	38-B
39	Integration Testing	B	Checks individual code components	39-D
40	System Testing	C	Validates the full system	40-C
41	System Integration Testing	D	Combines modules	41-A
42	Acceptance Testing	E	Final approval by users	42-E
43	Alpha Testing	F	Internal testing at developer's site	43-F

Group7: Dear learner: Choose the correct answer (9 questions):

No	Question	Answer
44	What is the primary advantage of using a <code>__construct</code> function? a) To destroy the object b) To automatically initialize an object's properties upon creation c) To set the access modifier d) To make properties private	b
45	Which access modifier restricts access to ONLY within the class itself? a) public b) final c) protected d) private	d
46	The OOP principle that promotes code reuse by allowing a class to inherit from another is called: a) Encapsulation b) Inheritance c) Polymorphism d) Abstraction	b
47	What keyword is used to create a new instance of a class (an object)? a) instance b) object c) new d) this	c
48	Inside a class method, which keyword refers to the current object? a) self b) that c) this d) object	c

49	Which principle restricts direct access to an object's data and uses methods to interact with it? a) Abstraction b) Inheritance c) Encapsulation d) Polymorphism	c
50	What is a class in OOP? a) An instance of an object b) A template for objects c) A built-in PHP function d) A type of variable	b
51	What will a __destruct function be called? a) When an object is created b) At the end of the script or when the object is destroyed c) Only when manually called d) At the beginning of the script	b
52	What does JSON stand for? a) Java Standard Object Notation b) JavaScript Object Notation c) JavaScript Operational Network d) Java Oriented Numbering	b

Group8: Dear learner: Complete the question (8 questions):

No	Question	Answer
53	To create a new object, you use the _____ keyword.	new
54	The _____ keyword refers to the current object inside a class method.	\$this
55	A constructor function starts with two underscores: _____.	__construct
56	A _____ property can only be accessed within the class itself.	private
57	In OOP, a class is a _____ for objects.	template
58	The _____ statement is used to choose between many conditions.	switch

59	To stop a loop completely, we use _____.	break
60	To skip one loop iteration, we use _____.	continue

Group9: Dear learner: Choose the correct answer (10 questions):

61	Who performs unit testing? a) Testers b) End-users c) Developers d) Business analysts	c
62	Which testing level focuses on individual code components? a) System Testing b) Unit Testing c) Acceptance Testing d) Integration Testing	b
63	What is the main purpose of System Integration Testing (SIT)? a) Test individual modules b) Verify interactions between different systems c) Check GUI functionality d) Only validate database queries	b
64	What is the goal of White-box testing? a) Test internal code and logic b) Verify only GUI elements c) Ignore error handling d) Only validate user stories	a
65	Which testing technique divides input data into equivalent partitions? a) Boundary Value Analysis b) Decision Table Testing c) Equivalence Partitioning d) State Transition Testing	c
66	What does BVA (Boundary Value Analysis) primarily test? a) Middle-range values	b

	b) Edge values of input ranges c) Only invalid inputs d) Database connections	
67	Which technique uses tables to represent input combinations and outputs? a) State Transition Testing b) Checklist-Based Testing c) Exploratory Testing d) Decision Table Testing	d
68	Which experience-based technique relies on tester intuition? a) Error Guessing b) Decision Table Testing c) Equivalence Partitioning d) Boundary Value Analysis	a
69	What is the main focus of Exploratory Testing? a) Following strict test scripts b) Unscripted, real-time testing c) Only validating acceptance criteria d) Ignoring edge cases	b
70	Which testing approach improves team communication? a) Black-box Testing b) White-box Testing c) Collaborative User Story Writing d) Error Guessing	c

Group10: Dear learner: Give The Correct Scientific Term (7 questions):

No	Question	Answer
71	This geometry manager places each widget in its own box and can stack them vertically or horizontally by default.	pack

72	This argument for the <code>pack</code> manager defines where the widget is placed in the parent widget, such as "n" for North or "s" for South.	anchor
73	This geometry manager arranges widgets within a matrix of rows and columns, like a spreadsheet.	grid
74	This geometry manager allows for absolute control over a widget's position using specific x and y coordinates.	place
75	This Tkinter widget allows you to display and edit multi-line text and supports embedded images and links.	Text widget
76	This method is used to add content to a Text widget, taking an index and the text characters as arguments.	insert()
77	The index format used by the Text widget's <code>get()</code> and <code>insert()</code> methods, which follows a 'line.column' structure.	index

Group11: Dear learner: Complete the question (8 questions):

No	Question	Answer
78	User interfaces are made up of multiple _____, arranged within a window.	widgets
79	You can set the text of a Label widget by using the _____ function.	config()
80	To align Label text to the left edge, you would set the anchor option to _____ or tk.W.	"w"
81	The _____ widget allows users to type input and can handle events like the Return key being pressed.	Entry
82	The actions which buttons perform are handled by Python _____.	functions

83	The root. _____() function is called to immediately close the application window.	destroy
84	The bd parameter of a Button sets the _____ width in pixels.	border
85	The askokcancel and askyesno functions are types of _____ boxes that ask for user confirmation.	message

Group12: Match CRUD Operations with Description (6 questions):

No	Column (A)	No.	Column (B)	Answer
86	INSERT	A	Retrieves data	86 → b
87	SELECT	B	Adds new data	87 → a
88	UPDATE	C	Removes records	88 → f
89	DELETE	D	Creates a new database	89 → c
90	CREATE DATABASE	E	Deletes a table	90 → d
91	DROP TABLE	F	Modifies existing data	91 → e

Exam 10

Group1: Dear learner: Answer the following questions. (5 questions).

1	Question	Name two major advantages of using PDO over the older <code>mysql_i</code> extension.
	Answer	1. works with many databases, not just MySQL. 2. Strong, consistent support for prepared statements to prevent SQL injection.

2	Question	Briefly describe the difference between <code>session_unset()</code> and <code>session_destroy()</code> .
	Answer	<code>session_unset()</code> removes all session variables, while <code>session_destroy()</code> destroys the

3	Question	What is the key difference between where cookies and sessions store their data?
	Answer	Cookies are stored on the client's browser, while sessions are stored on the server.

4	Question	What is the purpose of the session ID?
	Answer	It is a unique identifier stored on the client.

5	Question	How does <code>session_start()</code> work with cookies?
	Answer	It sends a Set-Cookie header with the session ID (PHPSESSID) to the client browser to maintain the session.

Group2: Dear learner: Give the Correct Scientific Term (7 questions):

No	Question	Answer
6	This geometry manager places each widget in its own box and can stack them vertically or horizontally by default.	pack

7	This argument for the <code>pack</code> manager defines where the widget is placed in the parent widget, such as "n" for North or "s" for South.	anchor
8	This geometry manager arranges widgets within a matrix of rows and columns, like a spreadsheet.	grid
9	This geometry manager allows for absolute control over a widget's position using specific x and y coordinates.	place
10	This Tkinter widget allows you to display and edit multi-line text and supports embedded images and links.	Text widget
11	This method is used to add content to a Text widget, taking an index and the text characters as arguments.	insert()
12	The index format used by the Text widget's <code>get()</code> and <code>insert()</code> methods, which follows a 'line.column' structure.	index

Group3: Match the Python GUI tool to its best use case (6 questions):

No	Function	No	Terms	Answer
13	Adds text to the widget.	A	<code>get()</code>	13-B
14	Retrieves text from the widget.	B	<code>insert()</code>	14-A
15	Removes text from the widget.	C	<code>image_create()</code>	15-D
16	Embeds an image into the widget.	D	<code>delete()</code>	16-C
17	Disables editing of the widget.	E	<code>tk.END</code>	17-F
18	The index representing the very end of the text content.	F	<code>config(state='disabled')</code>	18-E

Group4: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
19	SOLID principles are used in many design patterns.	True
20	The Open-Closed Principle states that a class should be open for extension and closed for modification.	True
21	Behavioral design patterns focus on object creation.	False
22	Structural design patterns are about organizing classes and objects into larger structures.	True
23	A single responsibility principle suggests that a class should have only one reason to change.	True
24	Creational patterns focus on class instantiation and object creation.	True
25	Design patterns speed up development by providing proven Solutions.	True
26	Design patterns are compared to Lego bricks because both help in building complex structures from smaller reusable units.	True
27	Creational design patterns are mainly concerned with how objects interact with each other.	False
28	Factory Pattern is used to ensure only one instance of a class exists.	False
29	Some design patterns can be combined to solve more complex problems.	True
30	The Singleton pattern ensures multiple instances of a class can exist at the same time	False

Group5: Dear learner: Choose the correct answer (7 questions).

No.	Question	Answer
31	Which of the following is NOT one of the SOLID principles? a) Single Responsibility Principle b) Open-Closed Principle c) Dependency Creation Principle d) Liskov Substitution Principle	C

32	Which design pattern category deals with communication between objects? a) Structural b) Behavioral c) Creational d) Functional	B
33	The main goal of the Interface Segregation Principle is: a) To increase inheritance b) To split large interfaces into smaller ones c) To combine interfaces d) To remove interfaces completely	B
34	Which of the following is the main task of design patterns? a) Provide high-level guidelines b) Provide reusable solutions to common design problems. c) Focus only on database logic d) Eliminate OOP concepts	B
35	Which of the following is a Creational Design Pattern? a) Adapter b) Singleton c) Composite d) Proxy	B
36	The main goal of Creational design patterns is: a) Code execution speed b) Object creation flexibility c) Database optimization d) User interface design	B
37	Structural patterns are mainly concerned with: a) Data storage b) Object creation c) Organizing relationships between objects d) Error handling	C

Group6: Give the Scientific Name Questions (6 questions).

No	Question	Answer
38	A set of five design principles that make object-oriented software more maintainable and extensible.	SOLID
39	A pattern that defines how objects communicate and distribute responsibilities.	Behavioral Design Pattern
40	A design pattern where the algorithm structure is defined in a base class with customizable steps.	Template Method Pattern
41	Ability to use existing code structure across different subclasses.	Reusability
42	Modifying specific steps while keeping the general algorithm intact.	Customization
43	Ability of code to adapt to changes in behavior.	Flexibility

Group7: Complete the Sentence Questions (10 questions).

No	Question	Answer
44	In design patterns classification, the two criteria are _____ and _____.	Purpose, Scope
45	The _____ Pattern allows you to define a family of algorithms and switch between them at runtime.	Strategy
46	A charger plug converter is an example of the _____ Pattern.	Adapter
47	The three main types of design patterns are _____, _____, and _____.	Creational, Structural, Behavioral
48	The Factory Pattern provides a common _____ for creating objects of different subclasses.	Interface
49	----- patterns focus on how objects are created .	Creational design

50	The ----- pattern helps to make incompatible interfaces compatible.	Adapter
51	----- pattern can improve performance by saving memory.	Singleton
52	----- patterns are concerned with relationships between objects.	Structural
53	The----- Method pattern provides a uniform interface for creating objects.	Factory

Group8: Dear learner: Answer the following questions. (8 questions).

54	Question	What does OOP stand for and name one advantage it has over procedural programming?
	Answer	Object-Oriented Programming. Advantages include: clearer structure, easier to maintain/modify/debug, code reuse (DRY), faster development.

55	Question	What is the difference between a class and an object?
	Answer	A class is a template or blueprint. An object is an instance of a class.

56	Question	What is the primary purpose of a constructor method (__construct)?
	Answer	To initialize an object's properties automatically when the object is created.

57	Question	What does the acronym DRY mean in the context of OOP advantages?
	Answer	Don't Repeat Yourself. It means code is reusable and not duplicated.

58	Question	Name the three access modifiers in PHP OOP.
	Answer	public, protected, private.

59	Question	What is the main benefit of the Inheritance principle?
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	Answer	It promotes code reuse by allowing a new class to inherit properties and methods from an existing class.
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60	Question	What does the \$this keyword represent inside a class method?
	Answer	It represents the current object instance calling the method.

61	Question	What are the two essential attributes an HTML form must have to allow file uploads?
	Answer	method="POST" and enctype="multipart/form-data".

Group9: Match String Functions with their Purpose (6 questions):

No	Functions	No.	Purpose	Answer
62	strlen()	A	Converts to lowercase	62→ c
63	strpos()	B	Replaces part of a string	63→ f
64	str_replace()	C	Returns length of a string	64→ b
65	substr()	D	Removes whitespace from start and end	65→ e
66	strtolower()	E	Extracts part of a string	66→ a
67	trim()	F	Finds first occurrence of substring	67→ d

Group10: Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
68	What is the primary focus of Agile methodology? a) Strict documentation b) Long development cycles c) Avoiding all changes d) Customer satisfaction and flexibility	d
69	Which Agile framework uses fixed-length sprints? a) Scrum b) Kanban	a

	c) Extreme Programming (XP) d) Waterfall	
70	What is the main purpose of a Sprint Retrospective? a) To demo new features b) To improve team processes c) To assign new tasks d) To write documentation	b
71	Which role removes obstacles in Scrum? a) Product Owner b) Scrum Master c) Developer d) Stakeholder	b
72	Kanban uses: a) A visual board (To Do, In Progress, Done) b) Fixed sprints c) Only written reports d) No task tracking	a
73	What does load testing simulate? a) A single API request b) Only manual testing c) Multiple users accessing an API d) Database queries	c
74	Which data formats are used in API testing? a) HTML and CSS b) XML and JSON c) PDF and DOCX d) MP3 and MP4	b
75	What is the main benefit of early bug detection in API testing? a) Saves time and money b) Increases costs c) Delays releases d) Requires more documentation	a
76	What does integration testing check? a) Single API components b) How different APIs work together	b

	c) Only security flaws d) UI responsiveness	
77	Which testing ensures an API meets its intended purpose? a) Performance testing b) Load testing c) Validation testing d) Security testing	c

Group11: Dear learner: Complete the question (8 questions):

No	Question	Answer
78	User interfaces are made up of multiple _____, arranged within a window.	widgets
79	You can set the text of a Label widget by using the _____ function.	config()
80	To align Label text to the left edge, you would set the anchor option to _____ or tk.W.	"w"
81	The _____ widget allows users to type input and can handle events like the Return key being pressed.	Entry
82	The actions which buttons perform are handled by Python _____.	functions
83	The root._____() function is called to immediately close the application window.	destroy
84	The bd parameter of a Button sets the _____ width in pixels.	border
85	The askokcancel and askyesno functions are types of _____ boxes that ask for user confirmation.	message

Group12: Dear learner: Complete the question (6 questions).

No	Question	Answer
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86	Testing software without predefined test cases is called _____ testing.	Exploratory
87	_____ testing checks individual code components in isolation.	Unit
88	_____ testing ensures different systems work together.	System Integration
89	_____ testing is done by real users before the final release.	Beta
90	A _____ describes a feature from a user's perspective.	User Story
91	_____ criteria define when a user story is complete.	Acceptance

Group13: Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
92	Agile testing starts only after development is complete.	False
93	Daily Scrums are optional in Agile teams.	False
94	Kanban uses fixed sprints like Scrum.	False
95	The Product Owner in Scrum sets task priorities.	True
96	API testing helps detect bugs early in development.	True
97	Security testing is not part of API testing.	False
98	Load testing simulates a single user accessing an API.	False
99	Unit testing checks how different APIs work together.	False
100	JSON and XML are common data formats in API testing.	True
101	API testing reduces costs by catching errors early.	True